Congruence carriers for Diophantine consistency equations

Thirteen positive unknowns in degree 34,
twelve positive unknowns in degree 118,
and two reductions over the integers

Research prepared for Vladimir Reshetnikov and the ProveIt repository
with OpenAI assistance

23 October 2026

Abstract

We give two fixed integer polynomials whose lack of strictly positive integer zeros is provably equivalent in ZFC to the syntactic consistency of ZFC. The first has thirteen unknowns, exact total degree 34, and a literal evaluator with 93 binary arithmetic operations. The second has twelve unknowns, exact degree 118, and 104 operations. Their coefficients are the complete, explicitly materialized compact native-compiler tuple already present in ProveIt.

The main construction puts divisibility information into two existing Pell square equations. The first equation simultaneously recovers a scale and a transport quotient, with positivity established before any Pell-index argument. The second carries two congruences in a prescribed order: an index is recovered modulo q^4, then a missing scale is recovered modulo q^5. A divided auxiliary norm removes one more positive unknown. We supply the signed index lemma, all new positivity arguments, and explicit canonical completeness constructions.

Separately, exact substitutions in a fixed forty-five-integer quartic source give equations in 37 unrestricted integer variables of degree 6, and 34 unrestricted integer variables of degree 8. Those two equations retain their source's different complete coefficient tuple. The archive includes literal equations, integer coefficient payloads, reconstruction programs, and exact algebraic checks. The proof composes ordinary mathematical reductions with named inherited compiler and refutation-calibration theorems. It is not a new end-to-end proof-assistant certificate, a consistency proof, or an optimality theorem.

Contents

1 Results, domains, and the comparison being made 3

2 The inherited source interface 4

3 Pell lemmas, including the signed auxiliary index 6

4 The thirteen-positive polynomial 8

5 Soundness of the thirteen-variable equation 9

6 Completeness and degree of the thirteen-variable equation 12

7 A divided norm gives twelve positive unknowns 14

8 Exact degree and cost of the twelve-variable equation 16

9 Two equations over unrestricted integers 17

10 Fixed coefficients and reproducible verification 20

1 Results, domains, and the comparison being made

Fix the recursive presentation of ZFC and the finite-proof coding used by the inherited ProveIt calibration. Write Con(ZFC) for the arithmetic assertion that this proof system has no finite syntactic refutation. The objective is a fixed polynomial with integer coefficients; there is no variable exponent, parameter selected by a consistency assumption, or uncounted existential register in any polynomial below.

Theorem 1.1 (Main constructions)**.** *There are explicitly specified polynomials P_{13} and P_{12}, with the complete coefficient tuple of Section 10, such that*

$$\mathsf{ZFC} \vdash (\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\boldsymbol{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}),$$
$$\mathsf{ZFC} \vdash (\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\boldsymbol{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}).$$

The exact degree and literal arithmetic-operation pairs are 34/93 *and* 118/104*, respectively. There are also explicitly specified polynomials Q_{37} and Q_{34} over unrestricted integers with the same consistency equivalence, in* 37 *variables of degree* 6 *and* 34 *variables of degree* 8*.*

The meaning of the displayed provability claim is ordinary mathematical provability formalizable in ZFC, using the inherited compiler and calibration theorem stated in Section 2. The archive does not export a newly checked object-language ZFC derivation or a complete Lean/Rocq build of that composition. The new polynomial identities, divisibility arguments, and existence reductions are proved here; the remaining source boundary is explicit.

1.1 A fixed, bounded comparison

All comparisons refer to ProveIt revision `beb0192cb8dc36f78c369c080f2adfbc1887a486`. The relevant source directories are 44, 57, 58, and 59 of `Computability/DiophantineConsistency/con-zfc-diophantine-forms`. The principal improvements and their costs are as follows.

Construction	Domain	Variables	Degree	Operations
Source 59	$\mathbb{Z}_{>0}$	13	126	91
P_{13}, this paper	$\mathbb{Z}_{>0}$	13	34	93
Source 57	$\mathbb{Z}_{>0}$	12	242	94
P_{12}, this paper	$\mathbb{Z}_{>0}$	12	118	104
Historical sextic	$\mathbb{Z}$	42	6	—
Q_{37}, this paper	$\mathbb{Z}$	37	6	—
Historical octic	$\mathbb{Z}$	36	8	—
Q_{34}, this paper	$\mathbb{Z}$	34	8	—

The positive-variable comparison is to the explicit source-57 and source-59 evaluators [6, 8]. A smaller degree need not give a smaller arithmetic circuit. The earlier thirteen-positive point of degree 132 at 83 operations, and twelve-positive point of degree 258 at 89 operations, retain lower operation counts [6]. The integer comparisons are to the preserved historical degree-six and degree-eight packets discussed in Section 9; the octic here uses two more residual equations than its historical comparator.

These are separate tradeoffs. In particular, a polynomial over $\mathbb{Z}_{>0}$ is not being advertised as a polynomial with the same number of unrestricted integer unknowns. Nor is a degree/variable comparison an operation-count comparison. Existing small-variable equations at much larger degrees remain different points in the catalogue.

1.2 Size conventions

Total degree counts the supplied unknowns and assigns degree zero to the four fixed coefficient ports. A named register denotes its fully substituted integer polynomial, not a new variable. The operation counts refer to the supplied straight-line graphs: every live binary addition, subtraction, and multiplication is charged, including multiplication by a small numeral. They are reproducible costs of those evaluators, not lower bounds on arithmetic complexity.

The large fixed integers are delivered as byte-exact payloads and by deterministic reconstruction programs. Their bit lengths are reported separately. A compact circuit does not make those coefficients small, and an expanded polynomial is not needed to specify them unambiguously.

2 The inherited source interface

The main positive-variable constructions use one whole authentic tuple

$$(\beta, K_0, M_C, M_F), \qquad B = 2^{590069}, \qquad \beta = B - 1.$$

Here M_F is the already shifted high-mask port: $M_F = M_F^{\mathrm{native}} + B - 1$. It must not be shifted a second time. The ports are not independently adjustable. The source-44 adapter proves its native interface for their joint values [1].

2.1 The normalized positive kernel

The old positive coordinates are

$$J, F, \alpha, z, f, h, i, T_{\mathrm{old}}, s, w, \tau, \eta, \zeta, y, \gamma_{\mathrm{old}}.$$

For this subsection only, define

$$\begin{aligned} &q = \beta J + 1, && C = q - F - \alpha, && X = wq, && Y = sq^3, \\ &E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y, \\ &H = 4a + 3, && \Delta = a^2 + H = (a+2)^2 - 1, && \ell = EkY, && I = ic^2. \end{aligned} \tag{1}$$

Further put

$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J, \tag{2}$$

$$D = X + ac + H\gamma_{\mathrm{old}},$$

$$V_{\mathrm{old}} = c(T_{\mathrm{old}} f - 1) - Rf^2. \tag{3}$$

The normalized positive source system is

$$\tau^2 - \ell(\ell + k) = 1, \qquad D^2 - \Delta c^2 = 1, \tag{4}$$

$$f^2 - \Delta I^2 = 1, \qquad \Delta(f^2 - 1)(V_{\mathrm{old}}^2 - y^2) + y^2 = 1, \tag{5}$$

$$k - hE - R = 1, \qquad (K_0 + w)C + (q - F) - z(q - 1) = 1. \tag{6}$$

Theorem 2.1 (Inherited compiler/calibration interface)**.** *For the whole compact source-44 tuple, a strictly positive solution of* (1)–(6) *yields a finite syntactic refutation of ZFC. Conversely, a finite refutation yields canonical strictly positive solutions of this system with the properties stated below.*

This theorem is the composition of the native-window/Pell compiler, its compact-coefficient adapter, and the literal blank-K215 halting/refutation calibration [1, 2, 3, 7]. The normalized strong coordinate used here is converted to the ordinary strong coordinate by $i_{\mathrm{ordinary}} = \Delta i$. Then

$$(i_{\mathrm{ordinary}} c^2)^2 = \Delta(f^2 - 1),$$

exactly matching the ordinary kernel. Thus the stated interface does not assume that a theorem for an arbitrary coefficient tuple applies to a different numerical instance.

2.2 Outer bounds available before index recovery

The authentic masks satisfy

$$0 < M_C, M_F^{\mathrm{native}} < B - 1, \qquad K_0 > 0.$$

Consequently, if $J, F, \alpha > 0$ and $C = q - F - \alpha > 0$, then

$$q \geq 16, \qquad 3q < R < q^4 - 2. \tag{7}$$

These inequalities use no divisibility assumption on directly supplied X, Y. Indeed, with $G = (q-1)(q-F) + \alpha$,

$$G = q\alpha + (q-1)C, \qquad q \leq G \leq q^2 - q,$$

and

$$0 < (M_C + qM_F)J < (q-1)(1+2q).$$

The resulting upper bound is $R < q^4 - q^3 + q^2 - 1 < q^4 - 2$; the lower bound is $R > q(q^2-1) > 3q$. The distinction matters in our construction: $C > 0$ is proved before (7) is used, whereas $q \mid X$ is recovered substantially later.

2.3 Canonical promises used in completeness

The source-44 padding controller and source-58 interface provide canonical witnesses with

$$q = B^N, \qquad J = 1 + B + \cdots + B^{N-1}, \quad N = 581197^e > 1, \tag{8}$$

$$R \equiv 3 \pmod 4, \quad X = 2^R, \qquad c = \psi_{a+2}(R), \tag{9}$$

$$\tau = \chi_{P_0}(n_0), \qquad k = 2\psi_{P_0}(n_0), \qquad P_0 = 2XY^2 + 1, \quad n_0 = (R+1)/2 \geq 3. \tag{10}$$

The Pell notation is defined in Section 3. In (8), N is odd, and e is a proof-only padding exponent. These are canonical completeness promises, not restrictions silently imposed on arbitrary solutions.

The compact adapter proves the controller at the actual final packed index, after its permitted dummy choices, and retains the low-mask identity $R \equiv 3 \pmod 4$ [1, Section 6]. We will also use

$$\gamma_{\mathrm{old}} > X \geq w. \tag{11}$$

Here is the source-58 growth proof in the present notation. Since $D = \sqrt{\Delta c^2 + 1} > (a+1)c$,

$$\gamma_{\mathrm{old}} = \frac{D - ac - X}{H} > \frac{c - X}{H}.$$

From $n_0 \geq 3$,

$$k \geq 2\psi_{P_0}(3) = 8P_0^2 - 2 > 32X^2Y^4, \qquad c > kY > 32X^2Y^5.$$

As $X \geq 16$, $Y \geq 1$, and $H = 4XY + 4Y + 3 \leq 9XY$, these inequalities give $c - X > XH$, proving (11). Merely knowing $\gamma_{\mathrm{old}} > 0$ would not justify the subtraction of w later.

3 Pell lemmas, including the signed auxiliary index

For an integer $A \geq 2$, put $\Delta = A^2 - 1$ and define

$$\chi_A(0) = 1, \quad \chi_A(1) = A, \qquad \psi_A(0) = 0, \quad \psi_A(1) = 1,$$

both sequences satisfying $z_{n+2} = 2Az_{n+1} - z_n$. Equivalently,

$$\chi_A(n) + \psi_A(n)\sqrt{\Delta} = (A + \sqrt{\Delta})^n.$$

We use their addition identities, positive-solution classification, and strong divisibility

$$\gcd(\psi_A(r), \psi_A(s)) = \psi_A(\gcd(r, s)).$$

These standard Pell facts and their use by the source are collected in the inherited audit [3]. The positive fundamental integer-coordinate solution for discriminant $A^2 - 1$ is $(A, 1)$.

Lemma 3.1 (Rational square-root integrality)**.** *If a rational number has integer square, it is an integer. In particular, if $d > 0$ and $z^2 = d^2 b$, with $z, b, d \in \mathbb{Z}$, then $d \mid z$.*

Proof. Write the rational number as a/b in lowest terms with $b > 0$. If a^2/b^2 is an integer, then $b^2 \mid a^2$, so coprimality forces $b = 1$. □

Lemma 3.2 (Signed abscissa step-down)**.** *Let $A \geq 2$, $m \geq 1$, and $0 < k < m$. If $t \geq 0$ and*

$$\chi_A(t) \equiv \chi_A(k) \pmod{\chi_A(m)},$$

then $t \equiv k$ or $-k \pmod{2m}$.

Proof. Write $F_m = \chi_A(m)$. The addition formulas give

$$\chi_A(j + 2m) \equiv -\chi_A(j), \qquad \chi_A(2m - j) \equiv -\chi_A(j) \pmod{F_m}.$$

Indeed $\chi_A(2m) = 2F_m^2 - 1$ and $\psi_A(2m) = 2F_m\psi_A(m)$. Reducing t modulo $2m$, and reflecting a representative above m, therefore gives a signed representative $\varepsilon\chi_A(s)$, $0 \leq s \leq m$. The case $s = m$ has residue zero, while $0 < \chi_A(k) < F_m$. If $s < m$ and $\varepsilon = -1$, the congruence would force

$$F_m \mid \chi_A(s) + \chi_A(k), \qquad 0 < \chi_A(s) + \chi_A(k) \leq 2\chi_A(m - 1) < F_m,$$

a contradiction. The last inequality follows from the recurrence for $A \geq 2$. For the positive sign, both representatives are strictly between zero and F_m; their equality modulo F_m is equality as integers. Strict growth gives $s = k$. The index operations used only sign and multiples of $2m$, proving the conclusion. □

This is the explicit signed argument of the ordinary source kernel [2, Section 5], also used by its normalized descendant [4, Section 5]. The hypothesis is $k < m$, not an unproved smaller half-interval for k.

Lemma 3.3 (Odd-index quotient identities)**.** *There are integer polynomials Q_v such that*

$$\begin{aligned} &\chi_Z(2v + 1) = ZQ_v(Z^2), \\ &\quad Q_v(0) = (-1)^v(2v + 1), \qquad\qquad Q_v(1 - A^2) = (-1)^v\psi_A(2v + 1). \end{aligned} \tag{12}$$

Moreover

$$\psi_Z(2v + 1) \equiv (-1)^v \pmod{Z^2}. \tag{13}$$

Proof. For the quotient polynomials,

$$Q_0(x) = 1, \quad Q_1(x) = 4x - 3, \quad Q_{v+1}(x) = (4x - 2)Q_v(x) - Q_{v-1}(x).$$

The odd subsequence of the Pell recurrence proves the first identity. Evaluation at zero proves the first evaluation by induction. At $x = 1 - A^2$, the recurrence coefficient changes sign relative to the odd ψ_A subsequence, whose initial values are $1, 4A^2 - 1$; this proves the second evaluation. For (13), write $b_v = \psi_Z(2v + 1)$. Then $b_0 = 1$, $b_1 = 4Z^2 - 1$, and $b_{v+1} = (4Z^2 - 2)b_v - b_{v-1}$. Modulo Z^2, induction gives $b_v = (-1)^v$. □

Lemma 3.4 (Signed auxiliary-index recovery)**.** *Let $A \geq 2$, $\Delta = A^2 - 1$, and let positive integers p, m, R, c, f, S satisfy*

$$c = \psi_A(p), \quad f = \chi_A(m), \quad c \mid m, \quad m > 2p, \quad S \geq 2, \quad c \mid S, \quad S^2 = \Delta(f^2 - 1),$$

and $0 < p, R < c/2$. If integers V, y satisfy

$$S^2V^2 - (S^2 - 1)y^2 = 1, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}, \tag{14}$$

then $p = R$. The signs of V, y are unrestricted.

Proof. The norm equation excludes $V = 0$ and $y = 0$. Positive Pell classification at parameter S gives $\nu \geq 1$ with

$$S|V| = \chi_S(\nu), \qquad |y| = \psi_S(\nu).$$

The index ν is odd: for even indices, $\chi_S(2v) \equiv (-1)^v \pmod{S}$, contrary to divisibility by S. Write $\nu = 2v + 1$. Then $|V| = Q_v(S^2)$. Since $S^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f}$, the first congruence in (14) and Lemma 3.3 give $\psi_A(\nu) \equiv \pm c \pmod{f}$. Squaring and using $\chi_A(2j) = 1 + 2\Delta\psi_A(j)^2$ gives

$$\chi_A(2\nu) \equiv \chi_A(2p) \pmod{\chi_A(m)}.$$

Lemma 3.2 applies with $k = 2p < m$, and yields $\nu \equiv \pm p \pmod{m}$, hence modulo c.

On the other hand, $c \mid S$ and the second evaluation in (12) at zero give $\nu \equiv \pm R \pmod{c}$. Thus $p \equiv R$ or $-R \pmod{c}$. The strict inequalities $0 < p, R < c/2$ exclude the negative identification and every nonzero multiple of c. Hence $p = R$. □

No main-root congruence, deleted index quotient, or positive sign of V is a hypothesis of this lemma. A signed strong root may be replaced by its absolute value as the modulus, with no change to its square or its congruences. The lemma is the exact coefficient-independent implication in [2, 6, Section 5], now stated without compiler notation.

Lemma 3.5 (Direct normalized rank)**.** *Suppose $c = \psi_A(p) > 2$ and positive integers f, I satisfy*

$$f^2 - \Delta I^2 = 1, \qquad c^2 \mid I.$$

Then $f = \chi_A(m)$, $I = \psi_A(m)$ for some $m \geq 1$, and

$$pc \mid m, \qquad c \mid m, \qquad m > 2p.$$

Proof. Pell classification gives m. Strong divisibility gives $p \mid m$. Write $m = pv$ and $D = \chi_A(p)$. Composition gives $I = c\psi_D(v)$. Since $D^2 \equiv 1 \pmod{c}$, the recurrence gives

$$\psi_D(v) \equiv vD^{v-1} \pmod{c}.$$

The number D is invertible modulo c. Since $c^2 \mid I$, we obtain $c \mid v$; hence $pc \mid m$, and $m \geq pc > 2p$. □

This argument is the normalized bridge in [3, Section 3.1] and [6, Section 5]. It needs no estimate $c > A\Delta^2$ from the older ordinary-strong rank lemma. Avoiding that estimate is essential when the first index quotient has not yet been restored.

Lemma 3.6 (Positive auxiliary reconstruction)**.** *Suppose $R \geq 3$, $R \equiv 3 \pmod 4$, $c = \psi_A(R)$, and*

$$f^2 - \Delta I^2 = 1, \qquad f, I > 0, \qquad c \mid I.$$

Set $S = \Delta I$, $Q = S^2$. There are positive integers T, u with

$$V = f^2(cT - R) - c, \qquad QV^2 - (Q-1)(V+u)^2 = 1.$$

They may be chosen with $y = V + u \equiv -1 \pmod Q$.

Proof. Write $R = 2v + 1$, where v is odd, and define

$$V_* = \frac{\chi_S(R)}{S} = Q_v(S^2), \qquad y_* = \psi_S(R).$$

They are positive integers satisfying

$$QV_*^2 - (Q-1)y_*^2 = 1.$$

Since $S^2 \equiv 1 - A^2 \pmod{f^2}$ and $c \mid S$, Lemma 3.3 gives

$$V_* \equiv -c \pmod{f^2}, \qquad V_* \equiv -R \pmod c, \qquad y_* \equiv -1 \pmod Q.$$

The strong equation gives $f^2 \equiv 1 \pmod c$, so $\gcd(c, f^2) = 1$. Therefore

$$T = \frac{V_* + c + Rf^2}{cf^2} \in \mathbb{Z}_{>0}. \tag{15}$$

It has $f^2(cT - R) - c = V_*$. Since $R \geq 3$, $S \geq 2$, we have $V_* \geq 4S^2 - 3 > 1$, and

$$y_*^2 - V_*^2 = \frac{V_*^2 - 1}{Q - 1} > 0.$$

Thus $u = y_* - V_* > 0$. This proves all claims. □

This is the squared-modulus reconstruction of source 51 and source 58 [5, 7]. It chooses fresh auxiliary witnesses; it does not assert that every old quotient was divisible by f, or that every arbitrary auxiliary ordinate has the displayed residue.

4 The thirteen-positive polynomial

Supply exactly the thirteen unknowns

$$j, F, \alpha, f, i, T, X, Y, r, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}. \tag{16}$$

All subsequent letters are polynomial abbreviations. In particular, X, Y are supplied directly and are not initially assumed to be divisible by any power of q. Define

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha, \\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, \\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H, \\
\ell &= EkY, & b_0 &= X + ac, & I &= ic^2.
\end{aligned} \tag{17}$$

$$
\begin{aligned}
R &= \big((q-1)(q-F)+\alpha\big)(q^2-1)+(M_C+qM_F)J,\\
r_0 &= k-R-1, \qquad U = q^5,\\
D_* &= Ub_0 + H(U\gamma + q^4X - r_0), \qquad (18)\\
K_a &= \Delta(f^2-1), \qquad V = f^2(cT-R)-c, \qquad (19)\\
W &= (q-1)k+1, \qquad M_0 = q^3(q-1),\\
B_t &= (qK_0+X)C - qF,\\
Z &= M_0(\ell+k+r)+(q-1)Y - q^3WB_t. \qquad (20)
\end{aligned}
$$

The four residuals are

$$
\begin{aligned}
A_0 &= Z^2 - M_0^2\big(\ell(\ell+k)+1\big), \qquad (21)\\
A_1 &= D_*^2 - U^2(1+\Delta c^2), \qquad (22)\\
A_a &= (V+u)^2 - K_a u(2V+u) - 1, \qquad (23)\\
A_s &= f^2 - \Delta I^2 - 1. \qquad (24)
\end{aligned}
$$

Our first fixed polynomial is

$$\boxed{P_{13} = A_0^2 + A_1^2 + A_a^2 + A_s^2.} \qquad (25)$$

Equations (17)–(25), together with the coefficient binding, are a literal division-free specification. The archive graph `positive/p13_degree34.json` gives its shared-register evaluation.

4.1 Why these two carriers are useful

An equation $z^2 = m^2 b$, with integer b, forces the rational root z/m to be integral. In A_0, the coprime factors q^3 and $q-1$ recover two different quotients. The sign of its recovered root is allowed to vary; the bound on that root forces the transport numerator to be positive, and therefore forces $C > 0$.

The main square A_1 then gives

$$q^5 \mid q^4X - (k-R-1).$$

Its reduction modulo q^4 is used first, after the auxiliary Pell index has been recovered. Only then is divisibility by q^5 used to obtain $q \mid X$. The two stages are necessary: $X = wq$ is unavailable during the preliminary index bounds. The normalized strong equation is retained so those bounds can be proved without the removed index quotient.

5 Soundness of the thirteen-variable equation

At a positive zero of P_{13}, every residual in (21)–(24) vanishes. We reconstruct the source witnesses in an order that establishes each denominator and positivity premise before it is used.

5.1 The first carrier proves positivity before index recovery

By Lemma 3.1, $A_0 = 0$ gives

$$\tau = \frac{Z}{M_0} \in \mathbb{Z}, \qquad \tau^2 = \ell(\ell+k)+1.$$

The integer τ may be negative. Modulo q^3, the divisibility $M_0 \mid Z$ gives $q^3 \mid (q-1)Y$, hence $q^3 \mid Y$. Modulo $q-1$, it gives $(q-1) \mid B_t$, because $q^3 \equiv W \equiv 1 \pmod{q-1}$. Define the proof-only integers

$$s = Y/q^3 > 0, \qquad t = B_t/(q-1).$$

Dividing the carrier by M_0 yields

$$\tau = \ell + k + r + s - Wt. \tag{26}$$

Since

$$(\ell + k)^2 - \big(\ell(\ell + k) + 1\big) = \ell k + k^2 - 1 > 0,$$

we have $|\tau| < \ell + k$. Thus

$$Wt = \ell + k + r + s - \tau > r + s > 0.$$

The register W is positive, so $t > 0$ and $B_t > 0$. If $C \le 0$, however,

$$B_t = (qK_0 + X)C - qF < 0,$$

which is impossible. We have proved $C > 0$ without using $q \mid X$ or any auxiliary index theorem. The bounds

$$3q < R < q^4 - 2$$

now follow from Section 2.2.

5.2 The second carrier gives a positive main root

As 590069 is odd, $B \equiv 2 \pmod 3$ and $\beta \equiv 1 \pmod 3$. The imposed $J = 3j + 1$ gives $q \equiv 2 \pmod 3$. Since $q^3 \mid Y$,

$$H = 4Y(X + 1) + 3 \equiv 3 \pmod q, \qquad \gcd(q, H) = 1.$$

The numerator D_* is positive before we know $q \mid X$. Indeed $r_0 < k$, $c > kY$, and

$$UaY > H$$

because $U = q^5$, $q \ge 16$, $Y \ge q^3$, and $H = 4a + 3 < 6a$. Therefore

$$D_* > Uac - Hk > k(UaY - H) > 0. \tag{27}$$

From $A_1 = 0$, Lemma 3.1 gives

$$D = D_*/U \in \mathbb{Z}_{>0}, \qquad D^2 - \Delta c^2 = 1. \tag{28}$$

Reducing the numerator modulo $U = q^5$, and cancelling H using $\gcd(H, q) = 1$, gives

$$q^5 \mid q^4 X - r_0. \tag{29}$$

5.3 Preliminary Pell bounds with directly supplied scales

Put $A = a + 2$ and $P_0 = 2XY^2 + 1$. Main Pell classification gives

$$D = \chi_A(p), \qquad c = \psi_A(p), \qquad p \ge 1.$$

For the first norm, writing $v = XY^2$ gives $\tau^2 - v(v + 1)k^2 = 1$. Its positive fundamental integer-coordinate unit is $(2v + 1, 2)$: ordinate 1 would require $v^2 + v + 1$ to be a square strictly between v^2 and $(v + 1)^2$, while ordinate 2 occurs. Consequently

$$|\tau| = \chi_{P_0}(n_0), \qquad k = 2\psi_{P_0}(n_0), \qquad n_0 \ge 1.$$

Since $c > 1$, $p \geq 2$. If $n_0 = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$. This contradicts

$$c \geq \psi_A(2) = 2A \geq 4Y + 4.$$

Thus $n_0 \geq 2$, and

$$k \geq 4P_0 > 8XY^2, \qquad c > kY > 8XY^3 \geq 8q^9 > 2R. \tag{30}$$

Here $X \geq 1$, not $X \geq q$, is sufficient. Also $P_0 > A$ and $c > k$; monotonicity in parameter and index yields $n_0 < p$. The recurrence, with $A \geq 3$, gives

$$c = \psi_A(p) \geq (2A - 1)^{p-1} \geq 5^{p-1} > 2p \qquad (p \geq 2). \tag{31}$$

Thus both half-interval hypotheses for Lemma 3.4 are available before index recovery.

5.4 Normalized rank and the exact main index

From $A_s = 0$, the normalized strong equation is

$$f^2 - \Delta I^2 = 1, \qquad I = ic^2.$$

Lemma 3.5 provides an index m with

$$f = \chi_A(m), \qquad I = \psi_A(m), \qquad c \mid m, \quad m > 2p.$$

Set $S = \Delta I$. Then $S^2 = \Delta(f^2 - 1) = K_a$, and $c \mid S$. With $y = |V + u|$, the auxiliary residual becomes

$$S^2V^2 - (S^2 - 1)y^2 = 1. \tag{32}$$

It excludes $y = 0$ and $V = 0$. The formula for V gives

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c},$$

the latter because $f^2 = 1 + \Delta i^2 c^4 \equiv 1 \pmod{c}$. Equations (30), (31), and the normalized rank conclusion verify every hypothesis of Lemma 3.4. We obtain

$$p = R. \tag{33}$$

No sign of V, and no restored value of h, was used in this step.

5.5 Ordered recovery of the index and scale

Reducing (29) modulo q^4 gives $q^4 \mid r_0$. Since $P_0 \equiv 1 \pmod{2XY^2}$, the first-Pell recurrence gives

$$k \equiv 2n_0 \pmod{4XY^2}. \tag{34}$$

The modulus is a multiple of q^4, because $q^3 \mid Y$. Consequently

$$q^4 \mid 2n_0 - R - 1.$$

But $1 \leq n_0 < p = R < q^4$, so $|2n_0 - R - 1| < q^4$. Hence

$$2n_0 = R + 1. \tag{35}$$

Now (34) implies that

$$r_0 = k - R - 1 = k - 2n_0$$

is a multiple of $4XY^2$, hence of q^5. It is positive because $n_0 \geq 2$ and $\psi_{P_0}(n_0) > n_0$. Returning to the stronger congruence (29), we get $q^5 \mid q^4X$, so $q \mid X$. The restored coordinates

$$w = X/q > 0, \qquad h = r_0/E > 0 \tag{36}$$

are integers; integrality of h follows from $E = XY \mid 4XY^2$. This proves the removed index equation exactly.

5.6 The main quotient and transport quotient are positive

Put

$$\gamma_{\mathrm{old}} = \gamma + w - r_0/U.$$

It is an integer, and (28) says

$$D = b_0 + H\gamma_{\mathrm{old}} = X + ac + H\gamma_{\mathrm{old}}.$$

Since $D > (a+1)c$ and (30) gives $c > kY > X$,

$$H\gamma_{\mathrm{old}} = D - ac - X > c - X > 0.$$

Thus $\gamma_{\mathrm{old}} > 0$.

We have already proved $B_t > 0$ and $(q-1) \mid B_t$. Now $X = qw$ gives

$$B_t = q((K_0 + w)C - F).$$

Since $\gcd(q, q-1) = 1$, there is a positive integer

$$z_{\mathrm{new}} = \frac{(K_0 + w)C - F}{q-1}.$$

Set $z = z_{\mathrm{new}} + 1 > 0$. Then

$$(K_0 + w)C + (q - F) - z(q-1) = 1,$$

the original transport condition. Finally set $T_{\mathrm{old}} = fT > 0$. Its source auxiliary expression is exactly

$$c(T_{\mathrm{old}} f - 1) - Rf^2 = f^2(cT - R) - c = V.$$

Together with $s = Y/q^3 > 0$, $\tau_{\mathrm{old}} = |\tau| > 0$, and $y = |V + u| > 0$, we have reconstructed every positive source coordinate. Theorem 2.1 gives a finite ZFC refutation.

6 Completeness and degree of the thirteen-variable equation

6.1 Constructing all new positive coordinates

Start from a finite refutation and choose the canonical source witnesses of Section 2.3. Keep the normalized i, f and the positive outer coordinates. Since $B \equiv -1 \pmod 3$ and N is odd,

$$J = 1 + B + \cdots + B^{N-1} \equiv 1 \pmod 3.$$

As $J > 1$, the new $j = (J-1)/3$ is a positive integer.

The canonical $X = 2^R$ and $R > 3q$, $q \geq 16$, give $w = X/q > q$. From the source transport equation,

$$z(q-1) = (K_0 + w)C + q - F - 1 > w + 1 > q,$$

where $C \geq 1$ and $q - F = C + \alpha \geq 2$. Thus $z \geq 2$. Define

$$t = q(z-1) \geq q, \qquad r = Wt + \tau - \ell - k - s, \quad s = Y/q^3. \tag{37}$$

The old transport identity gives $B_t = (q-1)t$. Since $\tau > \ell$ and $k > XY > s$,

$$r > qW - k - s > qW - 2k = \big(q(q-1) - 2\big)k + q > 0.$$

All terms in (37) are integers. Substitution gives $Z = M_0\tau$, so $A_0 = 0$.

The source first index is $n_0 = (R+1)/2$, so (34) makes $r_0 = k - R - 1$ a positive multiple of $4XY^2$. For the canonical scales this is a multiple of q^7, and therefore of q^5. Set

$$\gamma = \gamma_{\mathrm{old}} - w + r_0/q^5. \tag{38}$$

This is a positive integer by (11). Its substitution yields $D_* = q^5 D$, hence $A_1 = 0$.

Finally apply Lemma 3.6 with $S = \Delta I$. It supplies positive T, u, with $V = V_*$ and $V + u = y_*$. This changes no carrier coordinate. The strong and auxiliary residuals vanish, and all thirteen supplied coordinates are positive. This proves the reverse existence implication. Combining it with Section 5 and negating establishes the first statement of Theorem 1.1.

6.2 Exact cancellation and degree

Let $q_0 = 3\beta j$, $k_0 = \eta + \zeta$, and define the homogeneous forms

$$R_0 = q_0^3(q_0 - F), \qquad B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

For the first carrier, write

$$T_0 = Z - M_0\ell = M_0(k + r) + (q-1)Y - q^3 W B_t.$$

The exact identity

$$A_0 = 2M_0\ell T_0 + T_0^2 - M_0^2\ell k - M_0^2 \tag{39}$$

removes the apparent degree-sixteen terms. Its degree-fifteen part is

$$\mathrm{in}_{15}(A_0) = -2q_0^8 k_0^2 X Y^2 B_{t,0}. \tag{40}$$

For the main carrier put $T_1 = U\gamma + q^4 X - r_0$. The exact identity

$$\begin{aligned} A_1 = {} & U^2(2acX + X^2 - Hc^2 - 1) \\ & + 2U(ac + X)HT_1 + H^2 T_1^2 \end{aligned} \tag{41}$$

gives degree 17, with

$$\mathrm{in}_{17}(A_1) = 8q_0^{10} k_0 X^2 Y^3 \gamma. \tag{42}$$

The other residuals have degrees 13 and 14:

$$\mathrm{in}_{13}(A_a) = 2X^2 Y^2 f^4 u R_0, \qquad \mathrm{in}_{14}(A_s) = -i^2 k_0^4 X^2 Y^6.$$

Thus only A_1^2 reaches degree 34, and

$$\mathrm{in}_{34}(P_{13}) = 64(3\beta j)^{20}(\eta + \zeta)^2 X^4 Y^6 \gamma^2. \tag{43}$$

The coefficient of $j^{20} X^4 Y^6 \eta^2 \gamma^2$ is $64(3\beta)^{20} > 0$. This proves exact degree after specialization to the fixed coefficient tuple, not just a generic formal-port upper bound.

6.3 The literal evaluator

The supplied evaluator has 52 multiplications, 24 additions, and 17 subtractions, for 93 live binary operations. One saving is the exact factoring identity

$$T_{\mathrm{tr}} = (qK_0 + X)C + q(q - F - 1), \qquad T_{\mathrm{tr}} - q(q-1) = B_t.$$

Consequently the less economical numerator

$$M_0(\ell + k + qW + r) + (q-1)Y - q^3 W T_{\mathrm{tr}}$$

equals Z identically. The independent checker retains this unfactored expression when comparing to the emitted graph, so it checks the rewrite rather than repeating it.

Two further shared-register rewrites preserve the exact polynomial. The register $f^2 - 1$, already needed by K_a, is reused in $A_s = (f^2 - 1) - \Delta I^2$. For the auxiliary residual, the evaluator puts $y_0 = V + u$ and computes

$$A_a = y_0^2 - K_a(y_0^2 - V^2) - 1.$$

The identity $y_0^2 - V^2 = u(2V + u)$ shows that this is exactly (23). These are arithmetic-cost changes; the four residuals and their degree certificates are unchanged.

7 A divided norm gives twelve positive unknowns

Supply exactly

$$j, F, \alpha, i, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}. \tag{44}$$

Use the common registers (17)–(20), and A_0, A_1 from (21)–(22), with the single change

$$J = 6j + 1. \tag{45}$$

The expressions K_a, V, A_a, A_s of the thirteen-variable construction are replaced by the following explicit polynomial registers:

$$D_f = 1 + \Delta I^2, \qquad C_* = \Delta i c, \qquad M = \Delta i c^3,$$
$$Q = MC_* = (\Delta I)^2, \qquad d_* = Q - 1, \qquad V = D_f(cT - R) - c, \tag{46}$$
$$A_* = d_* M(n^2 + D_f) - 2d_* n + C_*(1 - V^2), \tag{47}$$
$$B_* = 2d_*(Mn - 1), \qquad N_* = A_*^2 - B_*^2 D_f. \tag{48}$$

The twelve-variable polynomial is

$$\boxed{P_{12} = (1 + A_0)(1 + A_1)(1 + N_*) - 1.} \tag{49}$$

It has 104 live binary operations. All powers in these definitions are fixed, and no division occurs in the polynomial.

7.1 The exact identity behind the divided norm

For a proof-only f satisfying $f^2 = D_f$, put $t = n - f$. Direct expansion gives

$$A_* - B_* f = d_* M t^2 - 2d_* t + C_*(1 - V^2), \tag{50}$$
$$M(A_* - B_* f) = d_*\big(M(n - f) - 1\big)^2 - QV^2 + 1. \tag{51}$$

For (51), multiply (50) by M, use $MC_* = Q$, and $d_* = Q - 1$. The relation is an exact polynomial identity after substituting $D_f = f^2$; it is not a numerical approximation or a formal assertion about independent square classes.

The factorization $Q = MC_*$, with $M = \Delta i c^3$, $C_* = \Delta i c$, distributes the degree between the two terms of A_*. The residue of the canonical auxiliary ordinate permits division by M in the completeness proof. In the actual polynomial, that quotient is encoded by the supplied coordinate n and the cleared identity above.

7.2 The product finalizer recovers the three equations

The integer β is odd, and $J = 6j + 1$ is odd, so q is even. Thus $M_0 = q^3(q-1)$ and $U = q^5$ are even. It follows that

$$A_0 \equiv Z^2 \pmod 4, \qquad A_1 \equiv D_*^2 \pmod 4.$$

Also B_* is even, hence $N_* \equiv A_*^2 \pmod 4$. At a zero of (49), each of its three integer factors is $+1$ or -1. A factor -1 would make its residual $-2 \equiv 2 \pmod 4$, impossible for a square modulo four. Therefore

$$P_{12} = 0 \quad \Longleftrightarrow \quad A_0 = A_1 = N_* = 0 \qquad \text{on } \mathbb{Z}_{>0}^{12}. \tag{52}$$

No nonnegativity of the individual residuals is asserted.

7.3 Every twelve-variable zero has a positive thirteen-variable lift

For every positive input, $c \geq 3$, $\Delta > 0$, $Q > 1$, $M > 1$, and $D_f > c + 1$. In particular

$$B_* = 2(Q-1)(Mn-1) > 0.$$

There is no zero-denominator branch. If $N_* = 0$, the rational number

$$f = A_*/B_*$$

has square D_f; by Lemma 3.1 it is a nonzero integer. Set

$$y = M(n - f) - 1.$$

Identity (51) gives

$$QV^2 - (Q-1)y^2 = 1. \tag{53}$$

Both f and y may have either sign, but a positive lift exists without invoking a new signed source theorem.

Indeed, let $b = cT - R \in \mathbb{Z}$. The unconditional identity $V = D_f b - c$ and $D_f > c + 1$ imply

$$b \leq 0 \implies V \leq -c < -1, \qquad b \geq 1 \implies V \geq D_f - c > 1.$$

Thus $|V| > 1$ on every positive input. Equation (53) now gives

$$(Q-1)(y^2 - V^2) = V^2 - 1 > 0,$$

so $|y| > |V|$. Define the three coordinates of a thirteen-variable lift by

$$j_{13} = 2j, \qquad f_{13} = |f|, \qquad u_{13} = |y| - V. \tag{54}$$

All are strictly positive. The identity $3j_{13} + 1 = 6j + 1$ makes every outer register identical. Since $f_{13}^2 = D_f$,

$$f_{13}^2 - \Delta I^2 = 1, \qquad \Delta(f_{13}^2 - 1) = Q, \qquad f_{13}^2(cT - R) - c = V.$$

Also $V + u_{13} = |y|$, so its auxiliary residual vanishes by (53). The two carriers already vanish by (52). We have constructed an actual positive zero of P_{13}, whose soundness was proved above. This proves soundness of P_{12}, including all signed-root branches.

7.4 Canonical completeness and the ordinate residue

Start with canonical source witnesses. The repunit J is odd because B is even, and $J \equiv 1 \pmod 3$ by the odd canonical exponent. Thus $J \equiv 1 \pmod 6$, and $j = (J-1)/6 > 0$ is an integer. Choose r, γ exactly as in (37) and (38); these give $A_0 = A_1 = 0$.

Use Lemma 3.6 to reselect the canonical auxiliary pair V_*, y_* and quotient T. It gives

$$y_* = \psi_{\Delta I}(R) \equiv -1 \pmod{(\Delta I)^2} = -1 \pmod Q.$$

Since $M \mid Q$, the number

$$t = (y_* + 1)/M$$

is a positive integer. Take

$$n = f + t > 0. \tag{55}$$

Now $y_* = M(n-f) - 1$, so the auxiliary Pell identity and (51) give $M(A_* - B_* f) = 0$. Because $M > 0$, $A_* = B_* f$, whence $N_* = 0$. All twelve supplied coordinates are positive, and (49) vanishes.

The residue assertion is an all-index consequence of Lemma 3.3; finite Pell examples are not used as its proof. It is needed only for the freshly chosen canonical auxiliary ordinate. Together with the positive lift (54), this completes the second equivalence in Theorem 1.1.

8 Exact degree and cost of the twelve-variable equation

For this section put

$$q_0 = 6\beta j, \quad k_0 = \eta + \zeta, \quad c_0 = k_0 Y, \quad \Delta_0 = X^2 Y^2, \quad R_0 = q_0^3(q_0 - F),$$

and

$$B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F, \qquad D_{f,0} = \Delta_0 i^2 c_0^4.$$

The carrier computations (40)–(42) still apply with this q_0. In particular their exact degrees are $15, 17$. The relevant degrees for the norm are

$$\deg D_f = 14, \quad \deg C_* = 7, \quad \deg M = 11, \quad \deg Q = \deg d_* = 18, \quad \deg V = 18.$$

The two degree-43 contributions to A_* are $d_* M D_f$ and $-C_* V^2$. All its other summands have degree at most 32. Their combined leading form is

$$\begin{aligned} A_{*,43} &= C_{*,0} D_{f,0}^2 (\Delta_0 c_0^2 - R_0^2) \\ &= i^5 k_0^9 X^6 Y^{15} (k_0^2 X^2 Y^4 - R_0^2). \end{aligned} \tag{56}$$

This is nonzero. Meanwhile $\deg B_* = 30$, so $\deg(B_*^2 D_f) = 74 < 86$. Hence

$$\deg N_* = 86, \qquad \mathrm{in}_{86}(N_*) = A_{*,43}^2.$$

The product finalizer therefore has exact degree $15 + 17 + 86 = 118$, with leading form

$$\boxed{\begin{aligned} \mathrm{in}_{118}(P_{12}) = \; & -16 q_0^{18} i^{10} k_0^{21} X^{15} Y^{35} \gamma B_{t,0} \\ & \times (k_0^2 X^2 Y^4 - R_0^2)^2. \end{aligned}} \tag{57}$$

For a coefficient independent of the other three fixed ports, take

$$j^{18} i^{10} \eta^{25} X^{20} Y^{43} \gamma \alpha. \tag{58}$$

Its coefficient is exactly

$$16(6\beta)^{18} > 0. \tag{59}$$

Indeed, the minimum j-power 18 forces the choice of $-X\alpha$ from $B_{t,0}$ and of $k_0^4 X^4 Y^8$ from the squared bracket. No specialization of K_0, M_C, M_F can cancel this coefficient.

The literal graph has 60 multiplications, 26 additions, and 18 subtractions, totaling 104. It shares $ic = i\,c$, using

$$C_* = \Delta(ic), \qquad D_f = 1 + (\Delta c^2)(ic)^2.$$

The register Δc^2 is already used by A_1. These identities save a multiplication without changing either polynomial. For an alternative finalizer, the same three residuals give

$$P_{12}^{\mathrm{sos}} = A_0^2 + A_1^2 + N_*^2. \tag{60}$$

It has the same positive zeros, exact degree 172, and 103 binary operations. Its highest homogeneous form is $A_{*,43}^4$. The one-operation saving therefore comes with a larger degree. The archive retains this companion as `positive/p12_sos_split2.json`.

9 Two equations over unrestricted integers

The integer constructions take a different route. They begin with the literal forty-five-integer quartic system Q_{45} of source 57 [9] and eliminate variables that occur in monic defining equations. They retain the source's entire *mixed-parity-unrounded-1064921* tuple, with $B = 2^{1064921}$. This is different from the compact tuple used by P_{13}, P_{12}; neither family borrows individual ports from the other.

9.1 The full source system

All variables in this section are unrestricted integers. Define

$$\mathrm{Pos}(x_1, x_2, x_3) = 1 + x_1^2 + x_2^2 + x_3(x_3 + 1). \tag{61}$$

This is always strictly positive and represents every positive integer. For the latter assertion, apply Legendre's three-square theorem to $4n - 3$, which is 1 modulo four and is not of the excluded form $4^a(8b + 7)$. One square is odd and the other two are even; writing them as $(2x_3 + 1)^2, (2x_1)^2, (2x_2)^2$ gives (61).

The source's thirty-two arithmetic coordinates are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta, u.$$

They are accompanied by the thirteen guard coordinates consisting of four triples $\boldsymbol{e}, \boldsymbol{z}, \boldsymbol{b}, \boldsymbol{d} \in \mathbb{Z}^3$ and one integer v_q. Set

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$

and the quadratic expression

$$\mathcal{R} = qN - qL - q^2 + F(q + L - N) + \alpha L + (M_C + qM_F)J.$$

The twenty-eight source residuals are the left sides of the following equations:

$$\begin{aligned}
L-(q^2-1)&=0, & N-q(L+1)&=0,\\
X-wq&=0, & a-E-(2v+1)N&=0,\\
E-X(a-E)&=0, & c-\eta-k(a-E)&=0,\\
P-E(c-\eta)-k&=0, & \tau^2-P(P-k)-1&=0,\\
D-X-ac-\gamma H&=0, & \Delta-a^2-H&=0,\\
C_2-c^2&=0, & D^2-\Delta C_2-1&=0,\\
k-hE-\mathcal{R}-1&=0, & (K_0+w)C-F-z(q-1)&=0,\\
F_2-f^2&=0, & V+c-fo&=0,\\
V-cj+k-hE-1&=0, & V^2+g-y^2&=0,\\
I-(2r+1)C_2&=0, & B_a-I^2&=0,\\
B_a-\Delta(F_2-1)&=0, & y^2-B_ag-1&=0,\\
w-u^2&=0, & \eta-\mathrm{Pos}(\boldsymbol{e})&=0,\\
\zeta-\mathrm{Pos}(\boldsymbol{z})&=0, & q-2v_q^2&=0,\\
F\alpha-\mathrm{Pos}(\boldsymbol{b})&=0, & FC-\mathrm{Pos}(\boldsymbol{d})&=0.
\end{aligned} \tag{62}$$

These are the literal source residuals, with renamed guard coordinates and harmless regrouping of terms. In particular, the transport coordinate here is the shifted quotient satisfying $(K_0+w)C = F+z(q-1)$. Under the first two equations, $\mathcal{R} = ((q-1)(q-F)+\alpha)(q^2-1)+(M_C+qM_F)J$. Every displayed residual has degree at most two.

The source theorem states that the sum of their squares has no integer zero if and only if $\mathrm{Con}(\mathsf{ZFC})$, provably in ZFC [9, 10]. Its domain proof uses the dyadic guard $q = 2v_q^2$ to force $J > 0$; the two product guards force $F, \alpha, C > 0$, since these three nonzero integers have the same sign and sum to $q > 0$. The inherited odd-scale and signed-root arguments then restore the unguarded source coordinates. In particular they give $X = 2^R$ and $w > F$, so the literal transport residual above implies $z > 0$. Canonical completeness supplies the dyadic guard from the source tuple's odd binary exponent. These unbounded source arguments remain inherited; the reductions below alter none of them.

9.2 The exact elimination principle

Lemma 9.1 (Unit elimination)**.** *Suppose an integer polynomial system contains*

$$x - A(\boldsymbol{y}) = 0$$

with $A \in \mathbb{Z}[\boldsymbol{y}]$, and other residuals $F_i(x, \boldsymbol{y})$. Deleting that defining residual and substituting $x = A(\boldsymbol{y})$ in every other residual gives a system whose integer solutions are in bijection with the original ones.

Proof. The maps are projection $(x, \boldsymbol{y}) \mapsto \boldsymbol{y}$ and reconstruction $\boldsymbol{y} \mapsto (A(\boldsymbol{y}), \boldsymbol{y})$. They are inverse on the two solution sets by direct substitution. No division, root extraction, sign premise, or nonvanishing condition is involved. The same assertion holds when the defining residual has the opposite sign. □

Iterating the lemma along a triangular list preserves the exact integer solution set up to a specified bijection. Replacing the remaining system by the sum of its squared residuals preserves its zeros over $\mathbb{Z}$.

9.3 The thirty-seven-variable sextic

In (62), eliminate the following eight coordinates using their displayed source definitions, substituting everywhere:

$$\begin{aligned} B_a &= I^2, \quad C_2 = c^2, & F_2 &= f^2, & N &= q(L+1), \\ w &= u^2, \quad X = D - ac - \gamma H, & \eta &= \mathrm{Pos}(\boldsymbol{e}), & \zeta &= \mathrm{Pos}(\boldsymbol{z}). \end{aligned} \tag{63}$$

The remaining 37 source variables and 20 residuals define Q_{37} as their sum of squares. Equations (62) and (63) are a complete polynomial specification; `integer/sextic37.json` lists the resulting residuals without requiring the reader to perform the substitutions.

Every residual now has degree at most three. The useful choice is the definition of X: the main-root equation expresses it quadratically, whereas combining $X = wq$ with $w = u^2$ would make it cubic. The other equation for X becomes

$$D - ac - \gamma H = u^2 q,$$

which is cubic, and the substituted relation $E = X(a - E)$ remains cubic. The substituted positivity polynomials are quadratic and enter the relevant source residuals with at most one additional linear factor.

For an exact lower bound, the residual

$$I - (2r+1)c^2$$

contains the only occurrence of r in the new system. Its square contributes coefficient 4 to r^2c^4, and no other square contributes to that monomial. Therefore

$$\deg Q_{37} = 6$$

on every fixed coefficient slice, including the authentic source tuple. The full symbolic expansion has 718 monomials when the four ports are retained as formal symbols. Lemma 9.1 gives the exact solution bijection with Q_{45}, proving the consistency equivalence.

9.4 The thirty-four-variable octic

For the second elimination, use the following triangular substitutions:

$$\begin{aligned} L &= q^2 - 1, & N &= q^3, & X &= wq, \\ D &= wq + ac + \gamma H, & \Delta &= a^2 + H, & C_2 &= c^2, \\ F_2 &= f^2, & B_a &= I^2, & g &= y^2 - V^2, \\ \eta &= \mathrm{Pos}(\boldsymbol{e}), & \zeta &= \mathrm{Pos}(\boldsymbol{z}). \end{aligned} \tag{64}$$

There are 34 remaining integer variables and 17 residuals, all of degree at most four. Define Q_{34} as their sum of squares. The explicit residual list is `integer/octic34.json`. For example, two surviving equations are

$$y^2 - I^2(y^2 - V^2) - 1 = 0, \qquad I^2 - (a^2 + 4a + 3)(f^2 - 1) = 0.$$

The second has coefficient -1 on a^2f^2; the other residuals have smaller degree in the pair a, f. Thus the coefficient of a^4f^4 in their sum of squares is exactly 1. It follows that $\deg Q_{34} = 8$, independently of the fixed ports. Its symbolic sum of squares has 814 monomials.

Again Lemma 9.1 gives an exact integer-solution bijection with Q_{45}. Together with the inherited source theorem, this proves the two integer assertions in Theorem 1.1. The source guards remain algebraically present after substitution; no positive coordinate has been discarded on the strength of its name.

9.5 Comparison and scope of the integer reductions

The historical sextic and octic retained in the coefficient dossier have respectively 42 variables and 21 residuals, and 36 variables and 15 residuals. Their exact degrees are 6 and 8. Thus the new sextic saves five integer variables, and the octic saves two. The latter has two more residual equations. These comparisons are between independently calibrated equations with matching domains and degrees; no direct solution bijection with those historical comparison polynomials is claimed.

The actual new bijections are to the fixed Q_{45} source. Its literal packet has SHA-256

`2b4d3d372fd899042eb3db9ee25e085193204640d28009807fde0778bc9528ac`.

The source and its coefficient dossier are included, so both the precise polynomial and the binding of its four fixed ports can be checked.

10 Fixed coefficients and reproducible verification

10.1 The complete compact binding for the positive equations

The main binding is `positive/ACTIVE_BINDING.json`; its source is the complete source-44 manifest `coefficients/compact44/COEFFICIENT_BINDING.json`. The numerical ports are the unsigned little-endian integers in the four decompressed files below. Explicitly, bytes $b_0, \ldots, b_{t-1}$ represent $\sum_{\nu=0}^{t-1} b_\nu 256^\nu$.

Port	File in `compact44/coefficients/`	Bits	Raw bytes	Gzip bytes
β	`beta.bin.gz`	590069	73759	108
K_0	`K.bin.gz`	592884	74111	8940
M_C	`MC.bin.gz`	590069	73759	193
M_F	`MF_source.bin.gz`	590070	73759	204
Total			295388	9445

The raw SHA-256 digests, in the same order, are:

```
0664a868cbfef9869ea05c40a1c445563f685ff857092594c1c1223bc4b21f9d
c6d523bd32af8dd0345db843cdadf6932bfcad186912a6d112314b60657c554e
6007cabd5f5228618e8fb613f386d8fc3692a24520d6e4cd72af0262aae1ade4
48333b0d30a4edeb8579f44286cb59806c5622cb42b367f5b5f9ee827711c737
```

The files and deterministic builder, rather than truncated printed digits, specify the complete fixed integers. These digests authenticate that specification. In particular $\beta = 2^{590069} - 1$. The four ports are bound together and the M_F payload already contains the source shift.

The retained builder `build_coefficients.py`, ordered `k215_transitions.json`, and adapter `PROOF.md` are in `coefficients/compact44/`. The compact native encoding uses radix 2^{67}, layout length 8807, and $d = 67 \cdot 8807 = 590069$, with padding prime 581197 [1]. Its finite coefficient audits check the implemented masks and carry/controller arithmetic. Their finite scope does not replace the unbounded native-language and calibration theorems.

10.2 The separate integer binding

For Q_{37}, Q_{34}, the authoritative file is `integer/coefficients_source/ACTIVE_BINDING.json`. Its ports have the following bit lengths:

port	β	K_0	M_C	M_F
bits	1064921	1065043	1064921	1064922.

Here $\beta = 2^{1064921} - 1$, with the already shifted high-mask port. The complete raw payloads occupy 532479 bytes; their supplied gzip forms occupy 10466 bytes. The dossier retains the package-34 coefficient builder, independent decoder, ordered Wang943 input, literal machine input, source proof, historical comparator packets, and source-transfer ledger.

The recorded reconstruction reproduces the raw payloads, gzip payloads, and full historical binding manifest exactly. This establishes a complete numerical binding for these equations; it does not transfer the compact source-44 tuple to an integer source whose domain theorem was proved for a different tuple.

10.3 What the exact checks establish

The main archive entry point is

`python3 verify_all.py`.

The individual builders and checks are retained under `positive/` and `integer/`. For the two positive-variable equations, the checks read literal graphs as data and compare them with independently specified formulas.

For P_{13}, the checker expands the four residuals exactly over the thirteen unknowns and four symbolic fixed ports. It verifies (39), (41), the exact residual degrees, the leading form (43), all live inputs, and the operation ledger. The supplied circuit is thereby tied to the mathematical specification, including its nonzero coefficient after actual-port specialization.

For P_{12}, the checker expands the prefix containing the two carriers and the norm's input registers. It checks the norm suffix as an exact identity over independent cut registers D_f, M, C_*, V, n, and the finalizer over independent A_0, A_1, N_*. These identities compose to an exact check of the whole graph. It separately verifies (50) and (51), the leading form (57), and the selected coefficient (59). No random evaluation is substituted for an identity.

For the two integer equations, the builder checks that each eliminated residual is literally plus or minus the claimed unit definition. It performs exact sparse substitution, expands the full sum of squares, and verifies the variable counts, degrees, liveness, and coefficients $[r^2c^4]Q_{37} = 4$ and $[a^4f^4]Q_{34} = 1$. The new checks use explicit exceptions rather than removable assertions and can also be run under `python3 -O`. Historical programs that require a normal Python invocation retain that requirement in their own documentation.

10.4 Mathematical and formal boundaries

There are three distinct kinds of evidence in the archive. Exact finite algebra verifies that the delivered formulas, literal evaluators, substitutions, degrees, and numerical ports agree. The mathematical proofs in this paper establish the new unbounded existence reductions, including every recovery and completeness branch. The inherited native compiler and syntactic calibration identify the old source equations with finite ZFC refutations.

The theorem follows by composing these mathematical implications and negating the resulting existential statements. No proof of Con(ZFC), no example of a refutation, and no assumption

deciding whether one exists enters the construction of a coefficient. The ordinary proofs are formalizable in ZFC, but this archive does not claim a newly compiled proof-assistant endpoint for the entire chain. Internal separately conducted reviews are research checks within this collaboration, not external human endorsements.

The inherited source material and its original notices are retained. The source licenses include ProveIt's MIT No Attribution terms and the retained Wade license; the native compiler and machine lineage remain attributed within their original scopes. The changes proved here are the combined congruence carriers, their recovery order and positive completeness construction, the divided norm specialization, and the two exact integer eliminations. No new universal variable bound or global minimum is asserted.

References

[1] ProveIt contributors, *Complement-packed native clauses: a smaller complete coefficient tuple*, source 44, native proof and complete numerical binding. `44-b4-sparse/native/PROOF.md`.

[2] ProveIt contributors, *Ordinary strong norms and signed witnesses for the blank computation*, especially Sections 4–7. `35-cancel12`, preserved `ORDINARY_BLANK_PROOF.md`.

[3] ProveIt contributors, *Pell dependency audit for the 67-operation blank-input polynomial*, especially Sections 3.1 and 4. `36-small-tuple/proofs/INHERITED_PELL_AUDIT.md`.

[4] ProveIt contributors, *Eleven positive variables: normalized Pell recovery*, especially Section 5. `38-batching11/eleven/ELEVEN_POSITIVE_PROOF.md`.

[5] ProveIt contributors, *Squared-modulus auxiliary reconstruction.* `51-sparse-arith/short/SQUARED_MODULUS_PROOF.md`.

[6] ProveIt contributors, *Two separate quadratic norms: twelve positive variables at degree 242*, especially Sections 5 and 9. `57-separated-degrees/low_degree/PROOF.md`.

[7] ProveIt contributors, *Integral-root carriers*, report, especially the inherited kernel and canonical completeness in Sections 4 and 6. `58-integral-root/report/report.tex`.

[8] ProveIt contributors, *Thirteen-positive degree-separated adjunct.* `59-degree-separated/positive13_adjunct/PROOF.md`.

[9] ProveIt contributors, *A quartic consistency equation in 45 unrestricted integer variables.* `57-separated-degrees/quartic/QUARTIC45_PROOF.md`.

[10] ProveIt contributors, *Odd-scale integer lifts and their domain reduction.* The historical proof and its exact source identities are retained in `integer/coefficients_source/low_degree/upstream/ODD_SCALE_LIFTS_PROOF.md`, with the coefficient dossier's transfer ledger and preserved comparison packets.

All linked catalogue references above are pinned to the comparison revision

`beb0192cb8dc36f78c369c080f2adfbc1887a486`.

Earlier commits appearing inside retained source documents describe their own historical dependency captures; they do not change the comparison pin or the active numerical bindings of this archive.